

Faculty of Applied Science and Technology**Lab #4 – Write Simple Script Files**

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1.

- 1.1) `% Question 1`
`% Problem 1: BMI Calculator`
`W = input('Enter your weight in lbs: ');`
`h = input('Enter your height in inches: ');`

`BMI = 703*W/h^2;`
`BMI = round(BMI, 1);`

`fprintf('The BMI is: %.1f\n', BMI);`

Enter your weight in lbs: 162
Enter your height in inches: 68
The BMI is: 24.6
- 1.2) `% Problem 2: Altitude Calculator`
`p = input('Enter the pressure in millibars: ');`
`h = 145366.45 * (1 - (p / 1013.25)^0.190289);`
`h = round(h);`
`fprintf('The altitude is: %d ft.\n', h);`

Enter the pressure in millibars: 394
The altitude is: 23915 ft.
- 1.3) `% Problem 3: Largest Sphere in Cone`
`R = input('Enter the base radius of the cone (in): ');`
`h = input('Enter the height of the cone (in): ');`
`r = (R * h) / (R + sqrt(R^2 + h^2));`
`r = round(r, 1);`
`fprintf('The radius of the largest sphere is %.1f in.\n', r);`

Enter the base radius of the cone (in): 8
Enter the height of the cone (in): 22
The radius of the largest sphere is 5.6 in.

1.4)

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% Problem 4: Radioactive Decay
t_half = input('Enter the half-life (years): ');
A0 = input('Enter the current amount (lb): ');
t = input('Enter the number of years in the future: ');
k = log(2) / t_half;
A = A0 * exp(-k * t);
fprintf('The amount after %d years is %.2f lb\n', t, A);
Enter the half-life (years): 24110
Enter the current amount (lb): 50
Enter the number of years in the future: 500
The amount after 500 years is 49.29 lb
```

1.5)

```
% Problem 5: Half Cylinder Volume
r = 14; % inches
L = 36; % inches
h = 0:2:14;
A = r^2 * acos((r - h)/r) - (r - h).*sqrt(2*r*h - h.^2);
V = A * L / 231; % convert to gallons
V = round(V, 1);
table(h', V', 'VariableNames', {'h_in', 'Volume_gal'})
```

h_in	Volume_gal
0	0
2	3
4	8.4
6	15.1
8	22.6
10	30.8
12	39.3
14	48

1.6)

```
% Problem 6: Garage Door Cable Force
theta = 0:10:90;
theta_rad = deg2rad(theta);
sin_alpha = (1 + 3*cos(theta_rad)) ./ sqrt((1 + 3*cos(theta_rad)).^2 + (3 - 3*sin(theta_rad)).^2);
alpha = asin(sin_alpha);
F = (300 * 4.5 .* sin(theta_rad)) ./ (3 * cos(alpha - theta_rad));
T = table(theta', F', 'VariableNames', {'Theta_deg', 'Force_lb'});
disp(T);
```

Theta_deg	Force_lb
0	0
10	116.59
20	209.31
30	283.11
40	341.08
50	385.26
60	417.07
70	437.57
80	447.84
90	450

2)

2.1)

```
% Problem 25: find coefficients a,b,c,d,e
X = [-2.5; -1.5; -0.5; 1; 3];
Y = [-62; -7.2; 8.3; 3.7; 45.7];
A = [X.^4, X.^3, X.^2, X, ones(size(X))];
p = A \ Y;
a = p(1); b = p(2); c = p(3); d = p(4); e = p(5);
fprintf('a = %.8f\nb = %.8f\nc = %.8f\nd = %.8f\ne = %.8f\n\n', a,b,c,d,e);
Yhat = A * p;
residuals = Y - Yhat;
disp(table(X, Y, Yhat, residuals))
fprintf('f(x) = (%g)x^4 + (%g)x^3 + (%g)x^2 + (%g)x + (%g)\n', a,b,c,d,e);

a = -0.05708514
b = 3.29258297
c = -4.00564214
d = -3.49760462
e = 7.96774892
```

X	Y	Yhat	residuals
-2.5	-62	-62	1.4211e-14
-1.5	-7.2	-7.2	3.5527e-15
-0.5	8.3	8.3	0
1	3.7	3.7	8.8818e-16
3	45.7	45.7	0

$$f(x) = (-0.0570851)x^4 + (3.29258)x^3 + (-4.00564)x^2 + (-3.4976)x + (7.96775)$$

2.2)

```
% Problem 27: Golfing
%coefficient matrix (each row = golfer)
A = [ 1 2 -1 -1;
      2 3 0 -1;
      1 4 -1 0;
      1 3 -2 0 ];
% Right-hand side (total points)
Y = [5; 12; 11; 8];
% Solve for [E (eagle); B (birdie); G (Bogey); D(DBogey)]
X = A \ Y;
E = X(1);
B = X(2);
G = X(3);
D = X(4);
fprintf('Points per Eagle: %.2f\n', E);
fprintf('Points per Birdie: %.2f\n', B);
fprintf('Points deducted per Bogey: %.2f\n', G);
fprintf('Points deducted per Double Bogey: %.2f\n', D);
```

Points per Eagle: 4.00
 Points per Birdie: 2.00
 Points deducted per Bogey: 1.00
 Points deducted per Double Bogey: 2.00

Tracing Exercise:

Let # be spaces for Answers

P: 0 4 8 12 16 20	#8.00
Q: 100 200 300 400 500 600	##400
r: 100 204 308 412 516 620	8.3516
fprintf ('%5.2f\n%5d\n8.3%d\n', p(3), q(4), r(5), q(2));	200
n1 = 5.45679	-87.35565
n2= 16.5673	-87.3556
n3 = 66	#5.45679
n4 = -87.355657	66
fprintf ('%8.5f\n', n4);	\$B##-87S
fprintf ('%4.4f\n', n4);	16.56300
fprintf ('%8.5f\n', n1);	16.57
fprintf ('%-12d\n', n3);	-87.36#71.4568
fprintf ('\$B%5.0fS\n', n4);	
fprintf ('%7.7f\n', n2);	
fprintf ('%3.2f\n%2.2f', n2,n4);	
fprintf ('%8.4f\n', n1+n3);	